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The Mechanism of How the Attractiveness of Low-Altitude Technology Presentation Influences Public Usage Intention: Based on An Extended Theory of Planned Behavior

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Abstract: With the rapid development of technologies such as unmanned aerial vehicles, low-altitude logistics, and urban air mobility, the low-altitude economy is increasingly emerging as a critical driver of industrial upgrading and urban governance transformation. However, as a representative emerging socio-technical system, the development of low-altitude technologies depends not only on technological maturity and institutional design but also heavily on public cognition, attitudes, and usage intention. Existing studies have predominantly focused on the technological feasibility and industrial models of the low-altitude economy, while paying relatively limited attention to the behavioral decision-making mechanisms of the public in low-altitude economic contexts. Against this backdrop, this study introduces the extended Theory of Planned Behavior to develop an analytical framework for public usage intention toward low-altitude technologies, systematically examining the formation pathways of usage intention from the interactive perspective of technology presentation attractiveness and psychological mechanisms. Specifically, the attractiveness of low-altitude technology presentation is conceptualized into four dimensions—visual attractiveness, appearance attractiveness, social attractiveness, and task attractiveness—and subjective norms, attitude, perceived behavioral control, and environmental indebtedness are incorporated as mediating variables; data are collected through a questionnaire survey and the hypotheses are empirically tested using structural equation modeling. The results indicate that different types of technology presentation attractiveness exert significantly differentiated effects on public usage intention. Specifically, visual attractiveness and appearance attractiveness significantly influence public attitudes, perceptions of behavioral control, and environmental responsibility cognition through multiple psychological pathways. Social attractiveness primarily operates through value identification and moral psychology, whereas the influence pathways of task attractiveness exhibit pronounced selectivity, mainly affecting subjective norms and environmental indebtedness. At the psychological mechanism level, attitude, perceived behavioral control, and environmental indebtedness all exert significant positive effects on public usage intention toward low-altitude technologies, with environmental indebtedness showing the strongest effect, while the direct effect of subjective norms does not reach conventional levels of statistical significance. Theoretically, this study extends the applicability of the Theory of Planned Behavior in the field of low-altitude economy research by revealing the critical roles of technology presentation and moral psychological mechanisms in emerging technology adoption; practically, it provides valuable implications for communication strategies, risk communication, and sustainable governance of low-altitude technologies.

Keywords: low-altitude economy; low-altitude technology usage intention; extended Theory of Planned Behavior; attractiveness of low-altitude technology presentation; environmental indebtedness.

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1 Introduction

In recent years, with the rapid development of technologies such as unmanned aerial vehicles, low-altitude logistics, and urban air mobility (UAM), the low-altitude economy has gradually emerged as a key force driving a new round of industrial transformation and urban governance transition [1]. Driven jointly by policy support, technological progress, and capital investment, application scenarios of the low-altitude economy continue to expand, and its development is expected not only to enhance operational efficiency and promote industrial upgrading but also to embody the core values of green, low-carbon, and sustainable development [2]. Against this backdrop, low-altitude technologies are entering the public sphere at an unprecedented pace, and public cognition, attitudes, and behavioral responses toward the low-altitude economy are increasingly becoming critical factors influencing its healthy development.

However, the rapid advancement of the low-altitude economy is accompanied by potential risks of irrationality in public decision-making processes [3]. On the one hand, low-altitude technologies are often presented in highly visualized, contextualized, and emotionally charged forms—such as drone swarm demonstrations, flying car test flights, and promotional displays of low-altitude application scenarios—which, while enhancing technological attractiveness, may also amplify emotional arousal and social influence effects [4]. On the other hand, when confronted with emerging technologies, the public often lacks sufficient information and experience in risk assessment, making their attitudes and behaviors more susceptible to external stimuli and social norms, thereby giving rise to rapid support or usage intentions based on intuition, emotion, or social pressure [5]. Such technology support or usage intentions formed without in-depth deliberation may to some extent obscure potential safety, ethical, and environmental issues, thereby posing latent risks to the sustainable development of the low-altitude economy.

Existing research on the low-altitude economy has primarily focused on technological feasibility, industrial models, and policy design, while systematic empirical investigation into the psychological mechanisms and behavioral logic of the public—particularly the formation of irrational usage intentions in low-altitude economic contexts—remains limited [6]. As a representative emerging socio-technical system, the development of the low-altitude economy is not merely a process of technological diffusion but also one in which public cognition, social norms, and value judgments are continuously reshaped [7]. Therefore, it is necessary to examine from a behavioral-theoretical perspective how modes of low-altitude technology presentation influence public attitude formation, perceived behavioral control, and moral psychological responses, thereby shaping usage intention.

Accordingly, this study introduces the extended Theory of Planned Behavior (TPB) to construct a theoretical model of public technology usage intention in the context of the low-altitude economy, and explores the pathways through which low-altitude technology presentation influences public behavioral decision-making from the interactive perspective of technological attractiveness and psychological mechanisms. Specifically, the study characterizes low-altitude technology presentation across dimensions such as visual attractiveness, appearance attractiveness, social attractiveness, and task attractiveness, integrates core TPB variables including subjective norms, attitude, and perceived behavioral control, and further incorporates environmental indebtedness as an extended construct to more comprehensively explain public usage intention and its potential irrational characteristics under the development of the low-altitude economy.

This study aims to uncover the psychological mechanisms underlying public behavioral decision-making in the low-altitude economy and to provide empirical evidence for understanding the social acceptance of emerging technologies. The findings not only enrich behavioral theory research in the field of the low-altitude economy but also offer decision-making references for governments and related enterprises in technology promotion, risk communication, and sustainable governance, thereby facilitating a more balanced development of the low-altitude economy between efficiency enhancement and social responsibility.

2 Theoretical Foundations and Development of Research Hypotheses

2.1 Theory of Planned Behavior and Technology Adoption Research

The Theory of Planned Behavior (TPB) is a prominent social psychological theory for explaining individual behavioral decision-making mechanisms, positing that actual behavior is primarily determined by behavioral intention, which in turn is jointly influenced by three key factors: attitude, subjective norm, and perceived behavioral control [8]. Specifically, attitude reflects an individual's overall evaluation of the outcomes of a given behavior, subjective norm captures the perceived expectations and pressures from significant others or social groups in social interactions, and perceived behavioral control represents the individual's subjective assessment of whether they possess the ability and resources required to perform the behavior [9].

Owing to its clear structure and strong explanatory power, the Theory of Planned Behavior has been widely applied in research on technology adoption, environmental behavior, and sustainable consumption. A substantial body of research indicates that when confronted with emerging technologies or complex decision-making contexts, individuals often do not rely solely on rational cost–benefit analyses, but instead determine whether to adopt a behavior through the combined effects of attitude formation, internalization of social norms, and perceptions of behavioral control [10]. Therefore, TPB provides a robust theoretical foundation for understanding public usage intention toward technologies in the context of the low-altitude economy.

However, the traditional Theory of Planned Behavior still exhibits certain limitations in explaining usage intention toward emerging technologies. On the one hand, TPB places greater emphasis on rational evaluative processes, while paying relatively limited attention to emotional arousal, moral psychology, and situational stimuli [11]. On the other hand, within highly visualized and socially embedded technology communication environments, public decision-making is often significantly shaped by modes of technology presentation and the surrounding social climate, which cannot be fully captured by the classic TPB variables alone [12]. Therefore, it is necessary to introduce extended constructs into the TPB framework to enhance the model's explanatory power for the low-altitude economy as an emerging socio-technical system.

2.2 Attractiveness of Low-Altitude Technology Presentation and Public Usage Intention

In the development of the low-altitude economy, low-altitude technologies typically do not enter the public sphere as abstract concepts, but are instead showcased through a variety of highly intuitive and contextualized forms, such as drone flight demonstrations, simulations of low-altitude logistics scenarios, flying car test flights, and multimedia promotional content [13]. While conveying information, these modes of technology presentation simultaneously shape the public's overall perceptions and psychological impressions of low-altitude technologies. Existing studies suggest that the attractiveness of technology presentation not only enhances public attention but may also influence attitudes and behavioral judgments through mechanisms of emotional arousal and cognitive simplification.

Drawing on relevant research, this study conceptualizes the attractiveness of low-altitude technology presentation into four dimensions: visual attractiveness, appearance attractiveness, social attractiveness, and task attractiveness. Visual attractiveness primarily reflects the visual effects and the level of sensory stimulation generated during the presentation of the technology [14]. Appearance attractiveness focuses on the degree of appeal of the technology or its carrier in terms of form design and external appearance. Social attractiveness emphasizes the symbolic meaning of the technology within social interactions and group identification. Task attractiveness reflects the extent to which the technology aligns with functional achievement, efficiency improvement, and practical application goals [14][15].

In the context of the low-altitude economy, these attractiveness factors not only help enhance public understanding of and interest in the technology, but may also influence psychological evaluation processes, thereby shaping attitude formation, perceptions of social norms, and perceived behavioral control, ultimately affecting technology usage intention. In this study, "technology presentation attractiveness" does not refer to the objective performance of the technology itself, but rather to a composite perception of attractiveness formed through visual cues, appearance, social symbolism, and task-related functionalities during the display, communication, and presentation of low-altitude technologies.

2.3 Attractiveness of Low-Altitude Technology Presentation and Extended TPB Variables

2.3.1 Visual Attractiveness and the Extended Theory of Planned Behavior

Visual information plays a significant role in emotional arousal and attentional guidance in the communication of emerging technologies [16]. Through dynamic demonstrations, scenario simulations, and highly immersive displays, low-altitude technologies enable the public to intuitively perceive their operational modes and application potential [17]. Such visual presentations not only help lower the cognitive barriers to understanding the technology but may also reinforce positive evaluations of its perceived value among the public.

Within the framework of the Theory of Planned Behavior, the visual attractiveness of technology may influence public behavioral decision-making through multiple pathways [18]. On the one hand, effective visual presentation helps foster positive public attitudes toward low-altitude technologies. On the other hand, visualized displays are often accompanied by social endorsement and policy legitimization, thereby strengthening perceived subjective norms among the public. Moreover, intuitively presented operational processes and application scenarios can enhance public confidence in their own ability to adopt the technology, thereby increasing perceived behavioral control. Meanwhile, against the backdrop in which the low-altitude economy is widely framed within narratives of "green" and "sustainable" development, visualized technology presentation may also evoke environmental responsibility awareness and a sense of environmental indebtedness among the public. Based on the above analysis, the following hypotheses are proposed:

H1: The visual attractiveness of low-altitude technologies has a significant effect on the extended Theory of Planned Behavior variables.

H1a: Visual attractiveness has a positive effect on subjective norms.

H1b: Visual attractiveness has a positive effect on attitude.

H1c: Visual attractiveness has a positive effect on perceived behavioral control.

H1d: Visual attractiveness has a positive effect on environmental indebtedness.

2.3.2 Appearance Attractiveness and the Extended Theory of Planned Behavior

Appearance attractiveness refers to the degree to which low-altitude technologies or their carriers appeal to the public through form design, overall configuration, and the visual impressions they convey [19]. In the context of the low-altitude economy, low-altitude technologies are often endowed with symbolic meanings such as "high-tech," "futuristic," and "advanced," and their appearance design serves not only functional purposes but also implicitly conveys value judgments about the technology and broader social-symbolic signals. When encountering low-altitude technologies, the public tends to rapidly form initial impressions based on their external appearance, and subsequently evaluate the technology's reliability, safety, and social legitimacy.

Prior research indicates that the appearance characteristics of technologies or products may influence individual decision-making processes through mechanisms of emotional arousal and cognitive heuristics [20]. On the one hand, an attractive appearance design may enhance public liking for and trust in the technology, thereby shaping attitude formation and value evaluation. On the other hand, in contexts where emerging technologies are not yet fully understood, appearance features may serve as important cues for the public to assess technological controllability and the degree of social acceptance. Therefore, appearance attractiveness operates not only at an affective level but may also influence public behavioral decision-making through multiple psychological mechanisms.

From the perspective of the extended Theory of Planned Behavior, the appearance attractiveness of low-altitude technologies may influence public usage intention by shaping attitudes, subjective norms, perceived behavioral control, and moral psychological responses. However, the specific pathways and directions of these effects within the low-altitude economy as an emerging socio-technical system remain to be empirically examined. Based on the above analysis, the following hypotheses are proposed:

H2: The appearance attractiveness of low-altitude technologies has a significant effect on the extended Theory of Planned Behavior variables.

H2a: Appearance attractiveness has a significant effect on subjective norms.

H2b: Appearance attractiveness has a significant effect on attitude.

H2c: Appearance attractiveness has a significant effect on perceived behavioral control.

H2d: Appearance attractiveness has a significant effect on environmental indebtedness.

2.3.3 Social Attractiveness and the Extended Theory of Planned Behavior

Social attractiveness refers to the value of social recognition and the symbolic meaning for groups that low-altitude technologies embody during social dissemination processes [21]. In the development of the low-altitude economy, low-altitude technologies are often framed as key symbols of national strategy, urban competitiveness, or industrial upgrading, and their promotion involves not only technological diffusion but also intensive construction of social discourse [22]. In this process, the public may form social perceptions of low-altitude technologies by observing others' attitudes and behaviors as well as prevailing directions of public opinion.

Social attractiveness in the adoption of emerging technologies typically operates through social norms and demonstration effects [23]. When the public perceives that low-altitude technologies are supported by public policy, highlighted by the media, or endorsed by others, they are more likely to develop conformity tendencies, which in turn shape their behavioral decision-making. However, excessively strong social attractiveness may simultaneously challenge individuals' capacity for autonomous judgment. Under social pressure or group expectations, individuals may pay less attention to their own behavioral control capabilities and risk assessment, thereby weakening rational decision-making processes.

Within the framework of the extended Theory of Planned Behavior, the effects of social attractiveness on different psychological variables may not be uniform. On the one hand, social attractiveness influences individuals' perceptions of the behavioral environment for technology adoption through social identification and the prevailing climate of public opinion. On the other hand, its effects on individual attitudes and moral psychological responses may be constrained by personal values and cognitive foundations. Therefore, it is necessary to empirically examine the specific pathways through which social attractiveness affects subjective norms, attitudes, perceived behavioral control, and environmental indebtedness in the context of the low-altitude economy. Based on the above analysis, the following hypotheses are proposed:

H3: The social attractiveness of low-altitude technologies has a significant effect on the extended Theory of Planned Behavior variables.

H3a: Social attractiveness has a significant effect on subjective norms.

H3b: Social attractiveness has a significant effect on attitude.

H3c: Social attractiveness has a negative effect on perceived behavioral control.

H3d: Social attractiveness has a significant effect on environmental indebtedness.

2.3.4 Task Attractiveness and the Extended Theory of Planned Behavior

Task attractiveness refers to the effectiveness and degree of fit with which low-altitude technologies meet practical needs, enhance operational efficiency, and achieve specific functional objectives [24]. In the context of the low-altitude economy, public evaluations of low-altitude technologies are closely linked to their performance in concrete application scenarios such as logistics distribution, public services, environmental monitoring, and emergency management [25]. Therefore, compared with symbolic or aesthetic appeal, task attractiveness is more directly associated with the practical value of the technology.

From a behavioral decision-making perspective, when the public perceives clear advantages of low-altitude technologies in functional performance and goal attainment, they are more likely to develop recognition of the technology's rationality and social necessity [26]. Such function-oriented cognition may not only shape individual attitudinal judgments but also influence perceived subjective norms by reinforcing social consensus and public discourse. At the same time, against the backdrop in which low-altitude technologies are endowed with roles in environmental governance and sustainable development, task attractiveness may also trigger moral reflection on environmental responsibility among the public.

However, in the context of the low-altitude economy—characterized by high technological complexity and uncertainty—it remains unclear whether task attractiveness can directly enhance public perceived behavioral control or whether it necessarily translates into positive attitudes. Therefore, it is necessary to empirically examine, within the extended Theory of Planned

Behavior framework, the pathways through which task attractiveness influences different psychological variables. Based on the above analysis, the following hypotheses are proposed:

H4: The task attractiveness of low-altitude technologies has a significant effect on the extended Theory of Planned Behavior variables.

H4a: Task attractiveness has a positive effect on subjective norms.

H4b: Task attractiveness has a significant effect on attitude.

H4c: Task attractiveness has a significant effect on perceived behavioral control.

H4d: Task attractiveness has a positive effect on environmental indebtedness.

2.4 The Mediating Role of Extended TPB Variables and Research Hypotheses

In the context of the low-altitude economy, public usage intention toward low-altitude technologies is not directly determined by technology presentation characteristics, but rather is gradually formed through a series of psychological cognitions, social judgments, and moral responses [27]. The extended Theory of Planned Behavior posits that external stimuli typically exert indirect effects on behavioral intention by shaping core psychological variables such as attitude, subjective norms, and perceived behavioral control. Meanwhile, within technology communication contexts that emphasize sustainable development, moral emotional factors may also play a significant role in behavioral decision-making.

2.4.1 The Mediating Role of Subjective Norms

Subjective norms reflect the expectations perceived by individuals from significant others or social groups regarding their behavior during the decision-making process [28]. In the development of the low-altitude economy, policy advocacy, media communication, and public opinion may jointly shape the social evaluative environment surrounding low-altitude technologies [29]. The attractiveness of low-altitude technology presentation may alter public normative judgments regarding whether the technology should be supported or used by influencing social consensus and group identification.

Therefore, subjective norms may play a mediating role between the attractiveness of low-altitude technology presentation and public usage intention toward low-altitude technologies, while the specific mechanisms and strength of this mediation remain to be empirically examined.

H5: Subjective norms may mediate the relationship between the attractiveness of low-altitude technology presentation and public usage intention toward low-altitude technologies.

2.4.2 The Mediating Role of Attitude

Attitude refers to an individual's overall evaluation of the outcomes of a given behavior and represents a key antecedent of behavioral intention. Through multiple dimensions of presentation attractiveness, low-altitude technologies may shape public integrated judgments of technological value, risk, and social significance at both cognitive and affective levels, thereby influencing usage intention [30]. Therefore, attitude may play a mediating role between the attractiveness of low-altitude technology presentation and public usage intention toward low-altitude technologies.

H6: Attitude mediates the relationship between the attractiveness of low-altitude technology presentation and public usage intention toward low-altitude technologies.

2.4.3 The Mediating Role of Perceived Behavioral Control

Perceived behavioral control reflects an individual's subjective assessment of whether they possess the abilities and resources required to perform a given behavior. In the context of the low-altitude economy, which is characterized by high technological complexity, modes of technology presentation may shape perceived behavioral control by influencing public perceptions of technological comprehensibility, operability, and safety [31]. Therefore, perceived behavioral control may serve as a mediating mechanism between the attractiveness of low-altitude technology presentation and public usage intention toward low-altitude technologies, exerting a substantial influence on usage intention.

H7: Perceived behavioral control mediates the relationship between the attractiveness of low-altitude technology presentation and public usage intention toward low-altitude technologies.

2.4.4 The Mediating Role of Environmental Indebtedness

Environmental indebtedness refers to the sense of moral responsibility that arises when individuals become aware of the negative environmental impacts caused by themselves or by societal development. Against the backdrop in which the low-altitude economy is framed as "green," "low-carbon," and "sustainable," the attractiveness of low-altitude technology presentation may evoke moral emotional responses among the public by reinforcing narratives of environmental governance [32]. Therefore, environmental indebtedness may play a mediating role between the attractiveness of low-altitude technology presentation and public usage intention toward low-altitude technologies.

H8: Environmental indebtedness mediates the relationship between the attractiveness of low-altitude technology presentation and public usage intention toward low-altitude technologies.

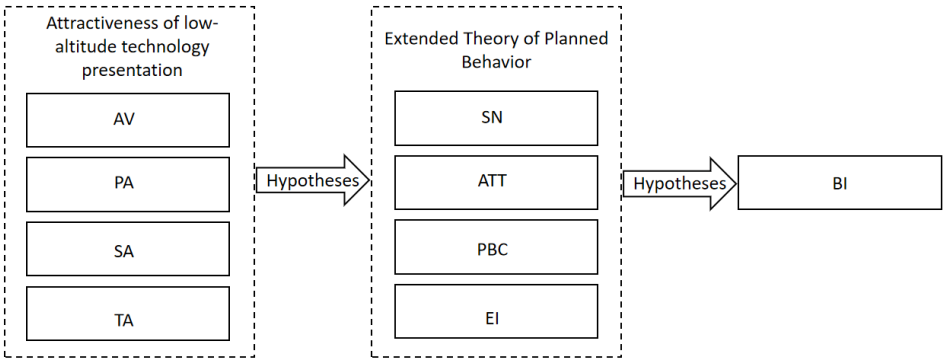

Figure1. Structural Equation Model Diagram

AV=Visual Attractiveness, PA=Appearance Attractiveness, SA=Social Attractiveness, TA=Task Attractiveness, SN=Subjective Norm, ATT=Attitude, PBC=Perceived Behavioral Control, EI=Environmental Guilt, BI=Behavioral Intention.

3 Research Design and Methodology

3.1 Development of the Research Model

Based on the preceding theoretical analysis and research hypotheses, this study adopts the extended Theory of Planned Behavior as the core analytical framework to construct a research model of public usage intention toward low-altitude technologies in the context of the low-altitude economy. From the interactive perspective of low-altitude technology presentation attractiveness and public psychological mechanisms, the model conceptualizes technology presentation attractiveness into four dimensions—visual attractiveness, appearance attractiveness, social attractiveness, and task attractiveness—and incorporates subjective norms, attitude, and perceived behavioral control as the core TPB variables, while introducing environmental indebtedness as an extended construct to systematically reveal how low-altitude technology presentation influences public usage intention through multiple psychological pathways.

In the research model, low-altitude technology presentation attractiveness is treated as an exogenous variable, the extended TPB variables are specified as mediating variables, and public usage intention toward low-altitude technologies is defined as the outcome variable. The model assumes that the attractiveness of low-altitude technology presentation initially shapes the public's psychological and cognitive structures and subsequently exerts indirect effects on public usage intention toward low-altitude technologies through mediating mechanisms such as subjective norms, attitude, perceived behavioral control, and environmental indebtedness. This research model facilitates a comprehensive understanding of the formation mechanisms underlying public technology usage intention in the context of the low-altitude economy.

3.2 Definition and Measurement of Variables

3.2.1 Attractiveness of Low-Altitude Technology Presentation

The attractiveness of low-altitude technology presentation refers to the overall perception of attractiveness formed by the public when exposed to information and display content related to low-altitude technologies. Drawing on prior research and

considering the contextual characteristics of the low-altitude economy, this study conceptualizes this construct into four dimensions:

(1) Visual attractiveness, which refers to the visual effects, dynamic demonstrations, and overall sensory impact presented during the display of low-altitude technologies; (2) Appearance attractiveness, which denotes the degree to which low-altitude technologies or their carriers appeal to the public in terms of form design, structural configuration, and overall image; (3) Social attractiveness, which captures the social recognition value, group symbolic meaning, and demonstration effects embodied by low-altitude technologies during social dissemination; (4) Task attractiveness, which reflects the effectiveness and degree of fit of low-altitude technologies in meeting practical needs, enhancing efficiency, and achieving specific functional objectives.

All four dimensions are measured using multi-item scales, with item wording adapted and refined from established scales in technology attractiveness and social commerce research to ensure theoretical alignment and contextual validity of the measurement instruments.

3.2.2 Extended Theory of Planned Behavior Variables

(1) Subjective Norms

Subjective norms refer to the expectations and pressures perceived by the public from significant others, social groups, or public opinion during the formation of usage intention toward low-altitude technologies. This variable is used to assess the perceived social approval of low-altitude technology usage intention under the influence of social interactions and the broader social environment.

(2) Attitude

Attitude refers to the public's overall evaluation of using low-altitude technologies, encompassing subjective judgments regarding technological value, rationality, and potential impacts. This variable captures the public's integrated attitudinal orientation toward low-altitude technologies at both rational and affective levels.

(3) Perceived Behavioral Control

Perceived behavioral control refers to the public's subjective perception of whether they possess the abilities, resources, and conditions required to adopt low-altitude technologies. In the context of the low-altitude economy, this variable primarily reflects public judgments regarding technological controllability, safety, and their own level of understanding.

(4) Environmental Indebtedness

Environmental indebtedness refers to the sense of moral responsibility and psychological pressure experienced by the public when becoming aware of the negative environmental impacts caused by societal development or individual behavior. Against the backdrop in which the low-altitude economy is widely framed as green and sustainable, this variable is used to capture the environmental moral motivation embedded in public usage intention toward low-altitude technologies.

3.2.3 Public Usage Intention Toward Low-Altitude Technologies

Public usage intention toward low-altitude technologies refers to the public's tendency to support, accept, or use low-altitude technologies in the context of the low-altitude economy. This variable captures the behavioral responses exhibited by the public toward low-altitude technologies even in the absence of sufficient information or experience, reflecting not only rational judgment but also susceptibility to contextual cues and modes of information presentation.

3.3 Questionnaire Design and Data Collection

3.3.1 Questionnaire Design

This study employs a structured questionnaire as the primary data collection instrument. The questionnaire consists of three main sections: the first section includes screening items designed to verify whether respondents have basic awareness of or prior exposure to low-altitude technologies or the low-altitude economy; the second section collects respondents' basic demographic information, including gender, age, educational attainment, occupation, and income level; and the third section contains measurement items for the research variables, covering the four dimensions of low-altitude technology presentation attractiveness, the extended Theory of Planned Behavior variables, and public usage intention toward low-altitude technologies. All measurement items are rated on a five-point Likert scale, where 1 indicates "strongly disagree" and 5 indicates "strongly agree." To reduce the potential impact of common method bias, the order of items across different variables was randomized

during questionnaire design.

3.3.2 Data Collection and Sample Description

This study targets the general public as survey respondents and collects data through an online questionnaire. This approach facilitates broader sample coverage while ensuring anonymity and randomness throughout the survey process. The questionnaire was distributed via online platforms, with respondents participating voluntarily and completing the survey independently. During data cleaning, returned questionnaires were rigorously screened, and those exhibiting obvious logical inconsistencies, incomplete responses, or abnormal completion times were excluded, resulting in a set of valid samples for subsequent analysis. The sample size and structure of this study meet the fundamental requirements for sample adequacy and data quality in structural equation modeling analyses.

3.4 Data Analysis Methods

To systematically test the research hypotheses, this study employs multiple statistical analysis methods to process and analyze the data. First, descriptive statistical methods are used to analyze sample characteristics in order to understand th e basic composition of the sample; second, reliability analysis and confirmatory factor analysis (CFA) are conducted to assess the reliability and validity of the measurement scales; based on these results, a structural equation model (SEM) is constructed to empirically test the path relationships and mediating effects within the research model. Structural equation modeling allows for the simultaneous examination of multiple variable relationships and the integrated assessment of causal paths among latent variables, making it well suited for the complex theoretical model developed in this study. All data analyses are conducted within statistical software environments to ensure the scientific rigor and robustness of the results. Confirmatory factor an alysis and structural equation modeling are primarily performed using AMOS 23, while descriptive statistics and reliability analysis are conducted using SPSS Statistics 26.

3.5 Chapter Summary

This chapter systematically presents the research design for examining public usage intention toward low-altitude technologies in the context of the low-altitude economy, focusing on the research model, variable measurement, questionnaire design, and data analysis methods. Through a rigorous research design and well-structured methodological procedures, this chapter lays a solid foundation for subsequent empirical analysis and hypothesis testing.

Table 1. Questionnaire

Variable	Code	Item	Source
Visual Attractiveness	AV1	The presentations related to low-altitude technologies provide a clear and intuitive depiction of their application modes.	McCroskey and McCain (1974) [33], Shen (2019) [34]
	AV2	The overall visual effects of the low-altitude technology presentations are of high quality.	
	AV3	The content of the low-altitude technology presentations is visually highly appealing.	
	AV4	The overall presentation environment created by the low-altitude technology displays makes me feel comfortable and positively acknowledged.	
Appearance Attractiveness	PA1	I believe that low-altitude technologies or their carriers are attractive in terms of appearance design.	
	PA2	I feel that the overall form of low-altitude technologies conveys a strong sense of modernity.	
	PA3	The form and structural design of low-altitude technologies leave a favorable impression on me.	
	PA4	The appearance design of low-altitude technologies leads me to perceive a high level of technological sophistication.	
	PA5	The overall visual image of low-altitude technologies gives me a positive first impression.	
Social Attractiveness	SA1	I feel that the development philosophy of low-altitude technologies aligns well with my personal values.	
	SA2	I believe that supporting the development of low-altitude technologies is consistent with my views on social progress.	
	SA3	I believe that supporting low-altitude technologies helps align with the mainstream trends of current societal development.	

Task Attractiveness	TA1	The presentation content of low-altitude technologies sufficiently meets my need for relevant information.	
	TA2	The presentation content of low-altitude technologies helps me understand their practical application value.	
	TA3	Low-altitude technologies are perceived as capable of effectively improving the operational efficiency of related tasks or services.	
	TA4	The application of low-altitude technologies helps better accomplish specific social or economic tasks.	
Subjective Norm	SN1	Most people around me would support the development of low-altitude technologies.	Ajzen, Icek (1991) [35]
	SN2	People who are important to me generally approve of the application of low-altitude technologies.	
	SN3	Social opinion and policy orientation influence my attitude toward low-altitude technologies.	
	SN4	In the social environment, supporting low-altitude technologies is considered a reasonable choice.	
Attitude	ATT1	Overall, I support the promotion and application of low-altitude technologies in society.	
	ATT2	I believe it is understandable that different people hold different attitudes toward low-altitude technologies.	
	ATT3	Even if some low-altitude technologies are controversial, I can still understand public support for them.	
	ATT4	Overall, I hold a relatively positive view toward the development of low-altitude technologies.	
Perceived Behavioral Control	PBC1	If I choose to do so, I am capable of supporting or accepting the application of low-altitude technologies.	
	PBC2	Whether I support low-altitude technologies mainly depends on my own judgment.	
	PBC3	Even in the absence of sufficient experience, I am able to make judgments about whether to use low-altitude technologies.	
	PBC4	Under the current conditions, I believe that I have the ability to evaluate the application of low-altitude technologies.	
Environmental Guilt	EI1	If social development continues to consume environmental resources without restoration, I would feel a sense of indebtedness.	Feng, Y. (2025) [36]
	EI2	When the environment is damaged, I believe that everyone bears a certain degree of responsibility.	
	EI3	If I do not support technological development that benefits environmental improvement, I would feel uneasy.	
	EI4	In the context of increasingly severe environmental problems, I reflect on the responsibilities that individuals and society bear toward the environment.	
	EI5	When environmental issues are neglected, I feel that society falls short at the moral level.	
Behavioral Intention	BI1	When exposed to information about low-altitude technologies, I sometimes support certain applications that I had not carefully considered beforehand.	Feng, Y. (2025) [37]
	BI2	During the process of learning about low-altitude technologies, I may develop support intentions for some applications that were not originally planned.	
	BI3	When I see demonstrations or promotional materials related to low-altitude technologies, I sometimes develop a strong intention to support or accept them.	

AV=Visual Attractiveness, PA=Appearance Attractiveness, SA=Social Attractiveness, TA=Task Attractiveness, SN=Subjective Norm, ATT=Attitude, PBC=Perceived Behavioral Control, EI=Environmental Guilt, BI=Behavioral Intention.

4 Empirical Results and Analysis

4.1 Descriptive Analysis of Sample Characteristics

A total of 456 valid questionnaires were collected in this study. The gender distribution of the sample is relatively balanced, with males accounting for 52% and females for 48%. In terms of age distribution, the sample is mainly concentrated in the 26–40 age range, with 34.9% aged 26–30 and 30.0% aged 31–40, together accounting for more than 60%, indicating that the respondents are predominantly young and middle-aged individuals with a certain level of social and technological exposure.

Respondents aged 41–50 and over 50 account for 20.0% and 10.1%, respectively, while the proportion of those aged 18–25 is relatively low (5.0%).

With regard to educational attainment, respondents with a bachelor's degree or above constitute a relatively high proportion of the sample, including 41.0% with a bachelor's degree, 25.0% with a master's degree, and 3.3% with a doctoral degree or above, indicating a generally high educational level that facilitates understanding and evaluation of emerging technologies such as low-altitude technologies. Respondents with a junior college education account for 28.5%, while those with a high school education or below represent a small proportion (2.2%).

In terms of income distribution, monthly income is mainly concentrated in the range of RMB 3,001–8,000, with 41.7% earning RMB 3,001–5,000 and 27.0% earning RMB 5,001–8,000, indicating that the overall income level of the sample is moderate. The proportions of respondents with monthly incomes of RMB 8,001–12,000 and above RMB 12,001 are 18.6% and 5.3%, respectively, while those earning below RMB 3,000 account for 7.5%.

Regarding occupational distribution, the sample covers a wide range of social roles, with students constituting the largest group (32.5%), followed by private-sector employees (25.0%) and public-sector employees, including civil servants and state-owned or public institution staff (24.3%), while freelancers and other occupations account for 9.0% and 9.2%, respectively, indicating a relatively diverse occupational structure.

With regard to the frequency of reading and acquiring information related to drones, most respondents demonstrate a certain level of exposure to low-altitude technology information. Specifically, the combined proportion of respondents who obtain information once or twice a month and once or twice a week reaches 63.1%, indicating that the sample is not generally unfamiliar with drone and low-altitude technology topics, but has developed a baseline of sustained attention. The proportion of respondents who acquire relevant information once or twice every six months or annually is 30.9%, suggesting that a substantial share of participants engage with low-altitude technology information on an intermittent basis. By contrast, only 5.9% of respondents report never paying attention to or reading information related to drones. These findings indicate that, at the cognitive level, the sample possesses a basic foundation of low-altitude technology knowledge, enabling respondents to form relatively rational attitudes and judgments toward such technologies, and thus is suitable for analyzing the mechanisms influencing public usage intention in the context of the low-altitude economy. Overall, the sample structure is reasonably sound and effectively captures public patterns of information exposure, cognitive formation, and behavioral tendencies in the low-altitude economy context, providing a reliable data foundation for subsequent empirical analyses.

Table 2. Descriptive statistics of population sample

Category	Description	Frequency	Percentage
Gender	Male	237	52
	Female	219	48
Age	18–24 years	23	5
	25–30 years	159	34.9
	31–40 years	137	30
	41–50 years	91	20
	51–60 years	46	10.1
Education	Junior High School or below	10	2.2
	High School/Vocational School	130	28.5
	Associate Degree	187	41
	Bachelor's Degree	114	25
	Master's Degree or above	15	3.3
Monthly Expenditure	Below RMB 3000	34	7.5
	RMB 3001–5000	190	41.7
	RMB 5001–8000	123	27
	RMB 8001–12,000	85	18.6
	Above RMB 12,001	24	5.3
Occupation	Civil servants / employees of state-owned enterprises / public institution staff	111	24.3

	Private-sector employees	114	25
	Students	148	32.5
	Self-employed individuals	41	9.0
	Other occupations	42	9.2
Frequency of browsing drone-related information	Once or twice a week	99	21.7
	Once or twice a month	189	41.4
	Once or twice every six months	114	25
	Once or twice a year	27	5.9
	Never	27	5.9
	Total	456	100

4.2 Reliability

Prior to conducting factor analysis, the suitability of the sample data was examined. The Kaiser–Meyer–Olkin (KMO) measure of sampling adequacy is 0.952, which is substantially higher than the recommended threshold of 0.90, indicating excellent suitability of the data for factor analysis. Meanwhile, Bartlett's test of sphericity yields an approximate chi-square value of 13,408.761 with 630 degrees of freedom and a significance level of $p < 0.001$, leading to the rejection of the null hypothesis that the correlation matrix is an identity matrix. These results indicate strong correlations among the measurement variables and confirm that the data structure is appropriate for subsequent factor analysis and structural equation modeling. Taken together, the KMO and Bartlett test results demonstrate a well-structured internal correlation pattern and sufficient shared variance among variables, satisfying the statistical prerequisites for conducting confirmatory factor analysis (CFA) and structural equation modeling (SEM). These findings further support the scientific rigor and stability of the subsequent model fit analyses.

Table 3. KMO Measure of Sampling Adequacy and Bartlett's Test of Sphericity Results

Construct	Number of Items
KMO Measure of Sampling Adequacy	.952
Bartlett's Test of Sphericity: Approximate Chi-Square	13408.761
Degrees of Freedom	630
Significance (p-value)	.000

To examine the internal consistency and measurement reliability of the scales, Cronbach's alpha coefficients were calculated for all latent variables. The results indicate that all latent variables exhibit Cronbach's alpha values exceeding 0.80, with visual attractiveness (AV) at 0.879, appearance attractiveness (PA) at 0.848, social attractiveness (SA) at 0.902, task attractiveness (TA) at 0.953, subjective norms (SN) at 0.893, attitude (ATT) at 0.953, perceived behavioral control (PBC) at 0.817, environmental indebtedness (EI) at 0.938, and usage intention (IB) at 0.840. The overall scale demonstrates a Cronbach's alpha coefficient of 0.957. According to established research standards, a Cronbach's alpha value greater than 0.70 indicates satisfactory internal consistency. These results suggest that all measurement scales employed in this study exhibit high reliability and are adequate for subsequent empirical analyses [38] [39].

Table 4. Cronbach's α Reliability Test Results for Each Latent Variable

Construct	Cronbach's Alpha	Number of Items	Construct	Cronbach's Alpha	Number of Items
AV	0.879	4	ATT	0.953	4
PA	0.848	5	PBC	0.817	4
SA	0.902	3	EI	0.938	5
TA	0.953	4	IB	0.840	3
SN	0.893	4	Total	0.957	36

AV=Visual Attractiveness, PA=Appearance Attractiveness, SA=Social Attractiveness, TA=Task Attractiveness, SN=Subjective Norm, ATT=Attitude, PBC=Perceived Behavioral Control, EI=Environmental Guilt, BI=Behavioral Intention.

To examine the underlying structural characteristics of the measurement scales, exploratory factor analysis (EFA) was conducted using principal component analysis [40]. The results indicate that a total of nine common factors were extracted based on the criterion of eigenvalues greater than 1. The cumulative variance explained by the extracted factors reaches 77.931%,

which is substantially higher than the commonly adopted 60% threshold in social science research, indicating that the extracted factors effectively capture the information structure of the original variables. Regarding the initial eigenvalues, the first factor has an eigenvalue of 15.080 and accounts for 41.889% of the variance, while the eigenvalues of subsequent factors decrease sequentially, suggesting the absence of a single-factor dominance problem in the scale structure. In the rotated sums of squared loadings, the variance explained by each factor is more evenly distributed, with individual factors accounting for between 6.211% and 10.320% of the variance, while the cumulative explained variance remains at 77.931%, further confirming the scale's structural stability and explanatory power [41]. In summary, the scales employed in this study meet the statistical requirements in terms of the number of extracted factors and cumulative variance explained, and are therefore suitable for subsequent confirmatory factor analysis and structural equation modeling.

Table 5. Total Variance Explained

Component	Initial Eigenvalues			Extraction Sums of Squared Loadings			Rotation Sums of Squared Loadings		
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	15.080	41.889	41.889	15.080	41.889	41.889	3.715	10.320	10.320
2	3.172	8.812	50.702	3.172	8.812	50.702	3.693	10.259	20.579
3	2.052	5.701	56.403	2.052	5.701	56.403	3.550	9.861	30.440
4	1.703	4.731	61.134	1.703	4.731	61.134	3.265	9.070	39.510
5	1.491	4.141	65.275	1.491	4.141	65.275	3.080	8.557	48.066
6	1.336	3.712	68.987	1.336	3.712	68.987	2.930	8.139	56.205
7	1.101	3.060	72.046	1.101	3.060	72.046	2.890	8.028	64.234
8	1.085	3.014	75.060	1.085	3.014	75.060	2.695	7.486	71.720
9	1.034	2.871	77.931	1.034	2.871	77.931	2.236	6.211	77.931
10	.595	1.654	79.585						
.....	.....	.....	.....						
36	.100	.278	100.000						

Extraction method: Principal Component Analysis.

After confirming the suitability of the data for factor analysis through the KMO measure and Bartlett's test of sphericity, this study further conducted exploratory factor analysis using principal component analysis combined with Kaiser-normalized varimax rotation. The results show that the rotation converged successfully after seven iterations, indicating good model stability [42]. The rotated component matrix reveals that all measurement items exhibit high loadings on their respective factors and low cross-loadings on non-corresponding factors, indicating a clear overall structure. Specifically, all items show factor loadings greater than 0.69 on their intended components, with most exceeding 0.75, suggesting that the items effectively capture the underlying constructs. Meanwhile, high discriminant validity is observed among factors, with no significant multiple cross-loading issues, demonstrating good structural validity of the scale. Moreover, the nine extracted common factors are consistent with the latent variable structure specified in the theoretical model, further validating the rationality of the scale design and the stability of the measurement structure. In summary, the results of the rotated component matrix indicate that the scales employed in this study exhibit a clear factor structure and strong explanatory power, making them suitable for subsequent confirmatory factor analysis and structural equation modeling.

Table 6. Rotated Component Matrix[a]

Item	1	2	3	4	5	6	7	8	9
AV1						.740			
AV2						.668			
AV3						.731			
AV4						.695			
PA1				.777					
PA2				.800					
PA3				.722					
PA4				.727					
PA5				.705					

SA1								.793	
SA2								.793	
SA3								.814	
TA1	.921								
TA2	.907								
TA3	.917								
TA4	.893								
SN1					.846				
SN2					.793				
SN3					.715				
SN4					.699				
ATT1		.783							
ATT2		.799							
ATT3		.796							
ATT4		.779							
PBC1							.753		
PBC2							.799		
PBC3							.744		
PBC4							.782		
EI1			.751						
EI2			.780						
EI3			.747						
EI4			.741						
EI5									
BI1									.695
BI2									.774
BI3									.747

Extraction Method: Principal Component Analysis. Rotation Method: Varimax with Kaiser Normalization. a. Rotation converged after five iterations.

AV=Visual Attractiveness, PA=Appearance Attractiveness, SA=Social Attractiveness, TA=Task Attractiveness, SN=Subjective Norm, ATT=Attitude, PBC=Perceived Behavioral Control, EI=Environmental Guilt, BI=Behavioral Intention.

4.3 Validity Analysis

4.3.1 Confirmatory Factor Analysis and Model Fit

To assess the overall fit of the research model, several commonly used goodness-of-fit indices are employed, including the chi-square/degrees of freedom ratio (CMIN/DF), normed fit index (NFI), Tucker–Lewis index (TLI), comparative fit index (CFI), root mean square error of approximation (RMSEA), and goodness-of-fit index (GFI) [43]. The results indicate that the CMIN/DF value is 2.068, which falls within the excellent fit range of 1–3. The values of NFI, TLI, and CFI are 0.962, 0.975, and 0.980, respectively, all well above the recommended threshold of 0.90. The RMSEA value is 0.048, which is below the stringent cutoff criterion of 0.05. The GFI value is 0.948, also indicating an excellent level of model fit. Overall, all goodness-of-fit indices meet or exceed the evaluation standards proposed in prior research, indicating that the measurement model constructed in this study demonstrates good overall fit and provides a reliable foundation for subsequent structural equation modeling analyses.

Table 7. Results of CFA Model Fit Indices

Item	CMIN/DF	NFI	TLI	CFI	RMSEA	GFI
Excellent Value	>1, <3	>0.9	>0.9	>0.9	<0.05	>0.9
Good Value	>3, <5	>0.8	>0.8	>0.8	<0.08	>0.8
Result	2.068	0.962	0.975	0.980	0.048	0.948

4.3.2 Convergent Validity and Composite Reliability

Building on the confirmatory factor analysis, this study further examines the convergent validity and composite reliability of each latent variable. The results show that the standardized factor loadings of all measurement items on their corresponding latent variables exceed 0.60, with most loadings above 0.70, indicating that the items adequately represent their underlying constructs. Regarding the average variance extracted (AVE), all latent variables exhibit AVE values of no less than 0.50, with the AVE values for visual attractiveness (AV), appearance attractiveness (PA), social attractiveness (SA), task attractiveness (TA), attitude (ATT), subjective norms (SN), perceived behavioral control (PBC), environmental indebtedness (EI), and usage intention (BI) ranging from 0.500 to 0.836, thereby meeting established criteria for convergent validity. Meanwhile, the composite reliability (CR) values of all latent variables exceed 0.70, ranging from a minimum of 0.798 to a maximum of 0.953, indicating strong internal consistency and measurement stability of the constructs. Taken together, these results suggest that the measurement scales used in this study satisfy statistical requirements for both convergent validity and composite reliability, providing a robust measurement foundation for subsequent structural equation modeling analyses; the calculation formulas are presented as follows [44].

$$AVE = \frac{\sum_{i=1}^{n} \lambda_i^2}{n} \qquad CR = \frac{(\sum_{i=1}^{n} \lambda_i)^2}{(\sum_{i=1}^{n} \lambda_i^2) + \sum_{i=1}^{n} \delta_i}$$

Table 8. Measurement Model Results: Factor Loadings, AVE, and CR.

Second-Order Variables		Variables	Estimate	AVE	CR
AV4	<---	AV	.790	0.640	0.876
AV3	<---	AV	.724		
AV2	<---	AV	.844		
AV1	<---	AV	.835		
PA5	<---	PA	.607	0.552	0.859
PA4	<---	PA	.748		
PA3	<---	PA	.715		
PA2	<---	PA	.786		
PA1	<---	PA	.839		
SA3	<---	SA	.844	0.753	0.902
SA2	<---	SA	.879		
SA1	<---	SA	.880		
TA4	<---	TA	.891	0.836	0.953
TA3	<---	TA	.915		
TA2	<---	TA	.920		
TA1	<---	TA	.930		
ATT1	<---	ATT	.912	0.789	0.937
ATT2	<---	ATT	.861		
ATT3	<---	ATT	.898		
ATT4	<---	ATT	.880		
SN1	<---	SN	.742	0.608	0.861
SN2	<---	SN	.744		
SN3	<---	SN	.833		
SN4	<---	SN	.796		
PBC4	<---	PBC	.670	0.500	0.798
PBC3	<---	PBC	.692		
PBC2	<---	PBC	.764		
PBC1	<---	PBC	.691		
EI5	<---	EI	.624	0.692	0.917
EI4	<---	EI	.857		

EI3	<---	EI	.903		
EI2	<---	EI	.886		
EI1	<---	EI	.857		
BI1	<---	BI	.782		
BI2	<---	BI	.755	0.572	0.800
BI3	<---	BI	.730		

AV=Visual Attractiveness, PA=Appearance Attractiveness, SA=Social Attractiveness, TA=Task Attractiveness, SN=Subjective Norm, ATT=Attitude, PBC=Perceived Behavioral Control, EI=Environmental Guilt, BI=Behavioral Intention.

4.3.3 Discriminant Validity Assessment

Building upon the assessment of convergent validity and composite reliability, this study further evaluates discriminant validity among latent variables using the Fornell–Larcker criterion. Specifically, the square root of the average variance extracted (AVE) for each latent variable is placed on the diagonal of the correlation matrix and compared with the inter-construct correlation coefficients [45]. The results indicate that the square roots of AVE for all latent variables are greater than their correlations with other constructs. For example, the square root of AVE for visual attractiveness (AV) is 0.800, which exceeds its correlations with appearance attractiveness (PA), social attractiveness (SA), and other constructs. Likewise, appearance attractiveness (PA), social attractiveness (SA), task attractiveness (TA), subjective norms (SN), attitude (ATT), perceived behavioral control (PBC), environmental indebtedness (EI), and usage intention (IB) all satisfy this criterion.

In addition, none of the inter-construct correlation coefficients exceed 0.85, suggesting the absence of serious conceptual overlap among the variables. Taken together, these findings indicate that the measurement model developed in this study demonstrates good discriminant validity, with latent variables being clearly distinguishable in statistical terms, thereby providing a reliable basis for subsequent structural equation modeling. Moreover, the absence of excessively high correlations ($r > 0.90$) in the correlation matrix further rules out potential risks of multicollinearity and construct redundancy.

Table 9. Discriminant Validity Test Results for Each Dimension of the Scale

	AV	PA	SA	TA	SN	ATT	PBC	EI	IB
AV	1								
PA	.615**	1							
SA	.574**	.511**	1						
TA	.271**	.232**	.282**	1					
SN	.528**	.493**	.481**	.346**	1				
ATT	.623**	.600**	.540**	.200**	.567**	1			
PBC	.425**	.372**	.331**	.196**	.332**	.328**	1		
EI	.669**	.668**	.604**	.356**	.595**	.713**	.395**	1	
IB	.552**	.579**	.486**	.320**	.462**	.562**	.332**	.619**	1
√AVE	0.800	0.743	0.868	0.914	0.888	0.780	0.707	0.832	0.756

** At the 0.01 level (two-tailed), the correlation is significant.

AV=Visual Attractiveness, PA=Appearance Attractiveness, SA=Social Attractiveness, TA=Task Attractiveness, SN=Subjective Norm, ATT=Attitude, PBC=Perceived Behavioral Control, EI=Environmental Guilt, BI=Behavioral Intention.

4.4 Descriptive Statistics and Correlation Analysis

As reported in the table, descriptive statistics and normality test results for all measurement items and latent variables are presented. In terms of overall means (M), the latent variables range from 3.08 to 3.76, indicating that respondents hold moderately positive attitudes toward low-altitude technology presentation attractiveness, the extended TPB-related variables, and usage intention. Among them, perceived behavioral control (PBC) exhibits the highest overall mean (M = 3.760), suggesting that respondents generally perceive a high level of personal control over whether to use low-altitude technologies. The overall means of social attractiveness (SA) and visual attractiveness (AV) are relatively lower, yet remain above the scale midpoint of 3, indicating moderately favorable public evaluations of social recognition and visual presentation associated with

low-altitude technologies.

With respect to dispersion, the overall standard deviations (SDs) of the latent variables range from 0.67 to 1.14, suggesting a reasonable degree of response variability without extreme concentration or excessive dispersion, and indicating a relatively stable sample structure. Regarding normality, the absolute values of skewness for all measurement items are below 1.0, and the absolute values of kurtosis are well below 3.0, meeting the normality criteria proposed by Kline (1998). These results indicate that the sample data satisfy the basic assumptions of approximate normality and are suitable for subsequent confirmatory factor analysis and structural equation modeling [46].

Table 10. Descriptive Statistics and Tests of Normality for Each Dimension

Variables	Second-Order Variables	Mean	Standard Deviation (SD)	Skewness	Kurtosis	Population Mean (M)	Population SD
AV1	3.24	.885	.782	-.209	.541	3.1656	0.74180
AV2	3.17	.864	.747	-.292	.778		
AV3	3.11	.834	.696	-.277	.768		
AV4	3.14	.881	.776	-.336	.555		
PA1	3.68	.792	.628	-.451	.818	3.6351	0.67077
PA2	3.61	.808	.653	-.495	.873		
PA3	3.56	.831	.691	-.608	.949		
PA4	3.57	.787	.619	-.534	.746		
PA5	3.76	1.015	1.031	-.761	.502		
SA1	3.11	.968	.936	-.259	.371	3.0848	0.87490
SA2	3.10	.963	.928	-.178	.327		
SA3	3.04	.938	.880	-.331	.524		
TA1	3.24	1.225	1.500	-.345	-.734	3.2231	1.14317
TA2	3.22	1.179	1.390	-.283	-.649		
TA3	3.21	1.253	1.571	-.354	-.749		
TA4	3.23	1.226	1.504	-.327	-.729		
SN1	3.56	1.092	1.192	-.609	.099	3.6310	0.95458
SN2	3.63	1.109	1.230	-.675	.026		
SN3	3.70	1.083	1.173	-.704	.201		
SN4	3.63	1.103	1.217	-.628	.042		
ATT1	3.54	1.076	1.159	-.619	.071	3.5554	1.00347
ATT2	3.52	1.077	1.160	-.581	.047		
ATT3	3.57	1.081	1.168	-.634	.102		
ATT4	3.59	1.053	1.109	-.618	.150		
PBC1	3.84	.803	.645	-.726	1.260	3.7604	0.67043
PBC2	3.77	.799	.639	-.676	1.129		
PBC3	3.72	.842	.709	-.669	.688		
PBC4	3.71	.891	.793	-.758	.682		
EI1	3.17	.999	.998	-.224	-.038	3.1825	0.89376
EI2	3.16	.986	.971	-.233	.052		
EI3	3.22	1.001	1.001	-.234	-.070		
EI4	3.16	1.015	1.030	-.168	-.089		
EI5	3.21	.993	.986	-.287	-.047		
BI1	3.31	.861	.742	-.060	.434	3.2646	0.74267
BI2	3.27	.844	.712	.074	.355		
BI3	3.21	.855	.732	-.237	.751		

AV=Visual Attractiveness, PA=Appearance Attractiveness, SA=Social Attractiveness, TA=Task Attractiveness,

SN=Subjective Norm, ATT=Attitude, PBC=Perceived Behavioral Control, EI=Environmental Guilt, BI=Behavioral Intention.

Prior to conducting structural equation modeling, this study examined potential multicollinearity among the explanatory variables. Using usage intention (BI) as the dependent variable, collinearity diagnostics were performed for all explanatory variables included in the final model. The results show that the condition indices for all dimensions do not exceed 30, with the maximum condition index being 28.208, which is below the commonly used threshold for severe multicollinearity. Further examination of variance proportions indicates that, within dimensions associated with higher condition indices, no two or more core explanatory variables simultaneously exhibit high variance proportions (commonly defined as above 0.50), suggesting the absence of strong linear dependence among the explanatory variables. Taken together, the results of the condition index and variance proportion tests indicate that the research model does not suffer from significant multicollinearity, and that the explanatory variables can independently influence usage intention, thereby satisfying the statistical assumptions required for subsequent structural equation modeling [47].

Table 11. Collinearity Diagnostics[a]

Model	Dimension	Eigenvalue	Condition Index	Variance Proportions (Constant)	AV	PA	SA	TA	SN	ATT	PBC	EI
1	1	8.716	1.000	.00	.00	.00	.00	.00	.00	.00	.00	.00
	2	.092	9.726	.00	.01	.00	.01	.83	.00	.03	.00	.00
	3	.056	12.463	.12	.00	.01	.05	.06	.01	.06	.12	.06
	4	.037	15.303	.00	.00	.00	.78	.00	.19	.08	.00	.01
	5	.030	16.934	.00	.02	.01	.11	.04	.78	.09	.00	.07
	6	.021	20.225	.06	.30	.00	.05	.02	.01	.62	.01	.18
	7	.019	21.520	.00	.64	.01	.00	.02	.00	.06	.06	.46
	8	.018	22.285	.15	.00	.31	.00	.00	.00	.06	.66	.00
	9	.011	28.208	.67	.03	.65	.00	.02	.00	.00	.15	.22

a. Dependent variable: BI. AV=Visual Attractiveness, PA=Appearance Attractiveness, SA=Social Attractiveness, TA=Task Attractiveness, SN=Subjective Norm, ATT=Attitude, PBC=Perceived Behavioral Control, EI=Environmental Guilt, BI=Behavioral Intention.

4.5 Structural Equation Modeling Analysis and Hypothesis Testing

In the quantitative analysis stage, this study employs structural equation modeling (SEM) to examine the path relationships among the proposed hypotheses. SEM is a statistical technique capable of simultaneously analyzing complex relationships among multiple variables, making it suitable for both exploratory and confirmatory research. Following established SEM modeling guidelines and model fit criteria proposed in prior literature, model estimation is conducted using AMOS software to ensure the significance of path coefficients and the adequacy of overall model fit [48].

4.5.1 Structural Equation Model Fit Assessment

To assess the overall fit of the structural equation model, multiple goodness-of-fit indices are employed, including the chi-square/degrees of freedom ratio (CMIN/DF), incremental fit index (IFI), Tucker–Lewis index (TLI), comparative fit index (CFI), root mean square error of approximation (RMSEA), and goodness-of-fit index (GFI). The results show that the CMIN/DF of the structural model is 2.687, which falls within the ideal range of 1–3, indicating a good balance between model complexity and sample data. The IFI, TLI, and CFI values are 0.927, 0.919, and 0.926, respectively, all exceeding the recommended threshold of 0.90. The RMSEA value is 0.061, which is below the acceptable cutoff of 0.08. The GFI value is 0.834, meeting the criterion for acceptable model fit.

Taken together, these fit indices indicate that the structural equation model demonstrates an overall good fit, with a reasonable model specification that adequately captures the structural relationships among latent variables, making it suitable for subsequent path and hypothesis testing. Given that the structural model introduces multiple path constraints beyond the measurement model, a slight decline in overall fit indices compared with the measurement model is expected; nevertheless, all indices remain within statistically acceptable ranges.

Table 12. Model Fit Indices for the Structural Equation Model (SEM)

Item	CMIN/DF	IFI	TLI	CFI	RMSEA	GFI
Excellent Value	>1, <3	>0.9	>0.9	>0.9	<0.05	>0.9
Good Value	>3, <5	>0.8	>0.8	>0.8	<0.08	>0.8
Result	2.687	0.927	0.919	0.926	0.061	0.834

4.5.2 Results of Path Analysis

Based on the path estimates derived from the structural equation model, the research hypotheses are systematically tested, with the detailed results reported in the table. Examination of the paths from technology presentation attractiveness to the extended TPB variables reveals significant differences in how different types of attractiveness influence public psychological mechanisms. First, visual attractiveness (AV) exerts significant positive effects on subjective norms (SN), attitude (ATT), perceived behavioral control (PBC), and environmental indebtedness (EI) ($\beta = 0.302–0.596$, $p < 0.001$), indicating that clear and intuitive presentations of low-altitude technologies strengthen social approval, value evaluation, perceived self-control, and environmental responsibility; thus, H1a, H1b, H1c, and H1d are supported. Second, appearance attractiveness (PA) also shows significant positive effects on subjective norms, attitude, perceived behavioral control, and environmental indebtedness ($\beta = 0.161–0.495$, $p < 0.05$), suggesting that appealing form design and overall image shape public cognition and judgment through multiple psychological pathways; accordingly, H2a, H2b, H2c, and H2d are supported.

With respect to social attractiveness (SA), significant positive effects are found on subjective norms ($\beta = 0.241$, $p < 0.001$), attitude ($\beta = 0.304$, $p < 0.001$), and environmental indebtedness ($\beta = 0.285$, $p < 0.001$), whereas its effect on perceived behavioral control is not statistically significant ($\beta = 0.084$, $p = 0.078$). These findings indicate that social attractiveness primarily operates through value judgments and moral cognition, with a relatively limited direct influence on perceived self-control; therefore, H3a, H3b, and H3d are supported, whereas H3c is not. In contrast, task attractiveness (TA) exerts significant positive effects only on subjective norms ($\beta = 0.128$, $p < 0.001$) and environmental indebtedness ($\beta = 0.114$, $p < 0.001$), while its effects on attitude ($\beta = -0.023$, $p = 0.482$) and perceived behavioral control ($\beta = 0.042$, $p = 0.138$) are not significant. This suggests that task-oriented information presentation functions mainly through social norms and responsibility awareness, with weaker direct effects on subjective evaluation and perceived control; thus, H4a and H4d are supported, whereas H4b and H4c are not.

Further examination of the effects of the extended TPB variables on public usage intention (BI) shows that attitude ($\beta = 0.165$, $p < 0.001$), perceived behavioral control ($\beta = 0.128$, $p < 0.05$), and environmental indebtedness ($\beta = 0.315$, $p < 0.001$) all have significant positive effects, with environmental indebtedness exerting the strongest influence; therefore, H6, H7, and H8 are supported. By contrast, the effect of subjective norms on usage intention does not reach the 0.05 significance level ($\beta = 0.091$, $p = 0.054$), indicating that decisions about using low-altitude technologies rely more on individual attitudes, perceived control, and moral responsibility than on social pressure alone; accordingly, H5 is not supported. In summary, the path analysis results from the structural equation model broadly support the proposed theoretical framework, confirming that the attractiveness of low-altitude technology presentation influences public usage intention through multiple psychological mechanisms, while also revealing heterogeneity in effect magnitude and direction across different attractiveness dimensions.

Table 13. SEM Model Path Hypothesis Testing Results

Hypotheses			Path Relation	Estimate	S.E.	C.R.	p
SN	<---	AV	.416	.072	5.814	***	Supported
ATT	<---	AV	.596	.075	7.942	***	Supported
PBC	<---	AV	.302	.062	4.863	***	Supported
EI	<---	AV	.481	.063	7.636	***	Supported
SN	<---	PA	.321	.078	4.128	***	Supported
ATT	<---	PA	.490	.082	5.991	***	Supported
PBC	<---	PA	.161	.066	2.444	.015	Supported
EI	<---	PA	.495	.070	7.097	***	Supported
SN	<---	SA	.241	.056	4.291	***	Supported

ATT	<---	SA	.304	.058	5.250	***	Supported
PBC	<---	SA	.084	.048	1.765	.078	Not Supported
EI	<---	SA	.285	.049	5.853	***	Supported
SN	<---	TA	.128	.034	3.713	***	Supported
ATT	<---	TA	-.023	.033	-.703	.482	Not Supported
PBC	<---	TA	.042	.028	1.484	.138	Not Supported
EI	<---	TA	.114	.028	4.115	***	Supported
BI	<---	PBC	.128	.057	2.250	.024	Supported
BI	<---	EI	.315	.050	6.272	***	Supported
BI	<---	SN	.091	.047	1.925	.054	Not Supported
BI	<---	ATT	.165	.044	3.784	***	Supported

$p < 0.001$: ***. "<---" represents the hypothesized structural path from the independent variable to the dependent variable.

AV=Visual Attractiveness, PA=Appearance Attractiveness, SA=Social Attractiveness, TA=Task Attractiveness, SN=Subjective Norm, ATT=Attitude, PBC=Perceived Behavioral Control, EI=Environmental Guilt, BI=Behavioral Intention.

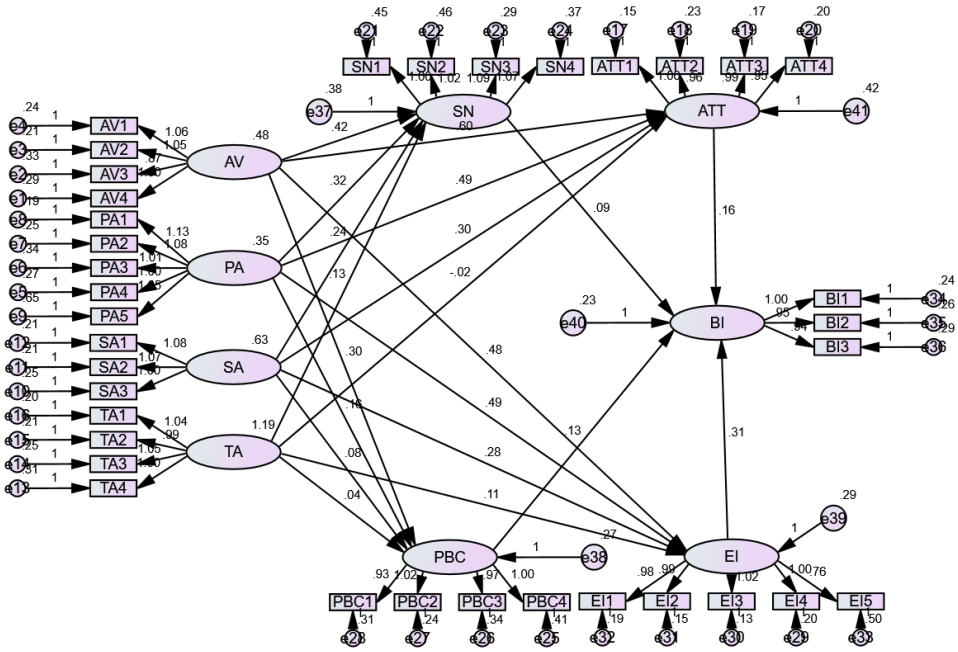

Figure 2. SEM Analysis Model

AV=Visual Attractiveness, PA=Appearance Attractiveness, SA=Social Attractiveness, TA=Task Attractiveness, SN=Subjective Norm, ATT=Attitude, PBC=Perceived Behavioral Control, EI=Environmental Guilt, BI=Behavioral Intention.

5 Discussion and Implications

5.1 Overall Discussion of the Research Findings

Based on the extended Theory of Planned Behavior, this study systematically examines the pathways through which technology presentation attractiveness influences public usage intention toward low-altitude technologies via multiple psychological mechanisms in the context of the low-altitude economy [49]. The empirical results demonstrate that different types of technology presentation attractiveness exert non-uniform effects on public psychological variables, exhibiting pronounced heterogeneity and selective influence patterns, which reflect the complexity of public cognition and decision-making associated with low-altitude technologies as an emerging sociotechnical system [50]. Overall, the extended Theory of Planned Behavior model shows strong explanatory power in the research context of the low-altitude economy. Attitude, perceived behavioral control, and environmental indebtedness all exert significant positive effects on public usage intention,

indicating that the adoption of low-altitude technologies is not driven solely by social pressure, but relies more heavily on individual value judgments, perceived self-control, and moral considerations regarding environmental responsibility.

5.2 Differentiated Mechanisms of Technology Presentation Attractiveness

The results indicate that visual attractiveness and appearance attractiveness exert significant positive effects on subjective norms, attitude, perceived behavioral control, and environmental indebtedness, demonstrating relatively comprehensive and stable psychological influence pathways [51]. This finding suggests that, in the context of the low-altitude economy, intuitive, vivid, and aesthetically appealing technology presentations not only enhance affective perceptions but also deeply shape rational judgments and moral evaluations. Because technologies such as drones, low-altitude logistics, and urban air mobility rely heavily on visual communication, the quality of technology presentation becomes a critical cue through which the public interprets their safety, controllability, and social value.

By contrast, social attractiveness primarily operates through value identification and moral cognition [23]. It exerts significant positive effects on subjective norms, attitude, and environmental indebtedness, but does not significantly enhance perceived behavioral control. This indicates that social identification mainly shapes value judgments regarding whether a technology is socially endorsed, rather than strengthening perceptions of one's ability to use it, highlighting the normative and symbolic nature of social attractiveness in emerging technology adoption.

The effects of task attractiveness, however, exhibit clear functional boundaries [26]. The empirical results show that task attractiveness has significant effects only on subjective norms and environmental indebtedness, while its effects on attitude and perceived behavioral control are not significant. This suggests that, in the context of the low-altitude economy, merely emphasizing efficiency, functionality, or task performance is insufficient to directly alter public subjective evaluations or perceived behavioral control [51]. Functional information tends to influence public attitudes indirectly through narratives of social legitimacy and responsibility, rather than directly driving usage decisions [52].

5.3 Discussion of the Effects of Extended TPB Variables on Usage Intention

Among the extended TPB variables, attitude, perceived behavioral control, and environmental indebtedness all exert significant positive effects on public usage intention toward low-altitude technologies, with environmental indebtedness showing the strongest effect. This result indicates that, in the context of the low-altitude economy's strong linkage to sustainability issues, moral psychological factors have become a critical driving force in shaping public technology adoption [53]. Notably, the effect of subjective norms on usage intention does not reach the conventional level of statistical significance. This finding does not negate the role of social norms; rather, it suggests that decisions regarding the use of low-altitude technologies rely more on internal judgments than on external pressures, reflecting a highly individualized and internalized decision-making pattern [1]. This result provides a novel empirical perspective for understanding the social acceptance of emerging and uncertain technologies.

5.4 Theoretical and Practical Implications

At the theoretical level, this study systematically incorporates technology presentation attractiveness into the extended Theory of Planned Behavior, thereby expanding the applicability of the theory to research on the low-altitude economy and emerging sociotechnical systems. In addition, by introducing environmental indebtedness, the study highlights the pivotal role of moral psychology in sustainable technology adoption. From a practical perspective, the findings suggest that the promotion of low-altitude technologies should not rely solely on functional messaging, but should instead emphasize the synergistic roles of visual design, image construction, and narratives of environmental responsibility. When advancing the development of the low-altitude economy, governments and enterprises should place greater emphasis on risk communication and value guidance in order to enhance public rational understanding and moral identification, thereby facilitating the sustainable adoption of low-altitude technologies.

6 Conclusions, Limitations, and Future Research Directions

6.1 Research Conclusions

Based on the extended Theory of Planned Behavior, this study develops and empirically validates a mechanism model explaining public usage intention toward low-altitude technologies in the context of the low-altitude economy. The findings indicate that different types of technology presentation attractiveness influence public decision-making through differentiated psychological pathways, with visual attractiveness and appearance attractiveness exerting broad and stable effects, whereas social attractiveness and task attractiveness display stronger context dependency. Moreover, attitude, perceived behavioral control, and environmental indebtedness emerge as the core psychological drivers of public usage intention, while the direct effect of subjective norms is relatively limited.

6.2 Research Limitations

Despite its theoretical and empirical contributions, this study has several limitations. First, the data are drawn from a single national context, and cultural and institutional differences may limit the generalizability of the findings [54]. Second, this study uses usage intention as the dependent variable and does not directly observe actual usage behavior. In addition, the model does not incorporate important factors such as risk perception and institutional trust that may influence the adoption of low-altitude technologies.

6.3 Future Research Directions

Future research may be extended in several directions. First, longitudinal data or experimental methods could be employed to dynamically examine the evolution of public cognition and attitudes toward low-altitude technologies. Second, cross-cultural comparative studies could be conducted to test the stability of the model paths across different institutional contexts. Third, theoretical perspectives such as risk perception, trust, or sociotechnical co-construction could be incorporated to provide a more comprehensive explanation of the social acceptance mechanisms of the low-altitude economy. Fourth, integrating actual usage behavior or policy intervention contexts could enhance the practical explanatory power and policy relevance of future studies.

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